

Introduction to Microeconomics
Lectures 3 & 4 Supply & Demand

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ECON101
Fall 2024

- Why do male soccer players earn more than female soccer players?
- Why do diamonds cost more than water?
- Why do Picasso's paintings sell for more than other artists?
- Is it the cost of production that determines prices, or is it willingness to pay that determines prices?
- Alfred Marshall, in *Principles of Economics*, 1890 was the first to explain clearly how both costs and willingness to pay interact to determine market prices.

Definition

The **market** for any good or service consists of all (actual or potential) buyers or sellers of that good or service.

The demand curve

The **demand curve** for a good or service tells us the total quantity of that good or service that buyers wish to buy at each price.

Q^D (price of the good ; determinants of demand)

↳ quantity demanded

given fixed determinants ; $p \mapsto Q^D$

Determinants of Demand

$Q^D (p ; DET)$

examples

- tastes
- prices of substitutes / complements
- Income.
- Expectations of future prices
- # Consumers in market

Graphical representation of the demand curve

$$Q^D(p; D, \theta)$$

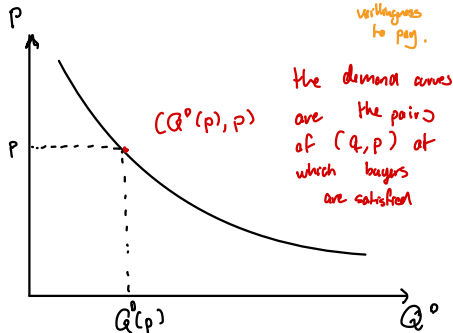
price is exogenous

Q^D endogenous variable.

Graph the inverse demand

$$(Q^D)^{-1}(q) \mapsto p$$

willingness to pay.



The demand curves are the pairs of (q, p) at which buyers are satisfied

Why is the demand downward sloping?

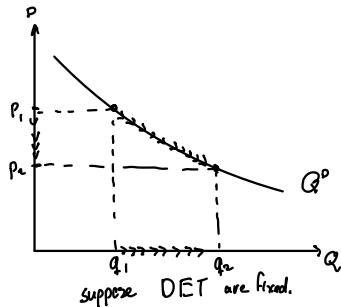
1. As the good becomes more expensive consumers can't afford as much of it
2. As the good becomes more expensive, you substitute towards cheaper alternatives

Law of Demand.

Horizontal interpretation: at price P , how much q is demanded

Vertical interpretation: given quantity q , what is the max. willingness to pay of the consumer?

Movements along the demand curve and shifts of the demand schedule

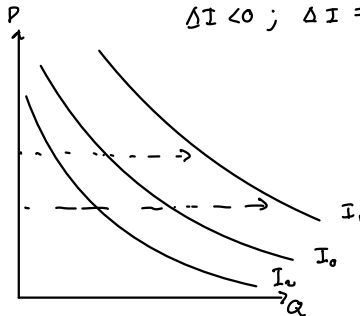


movement along the demand curve due to a price change.

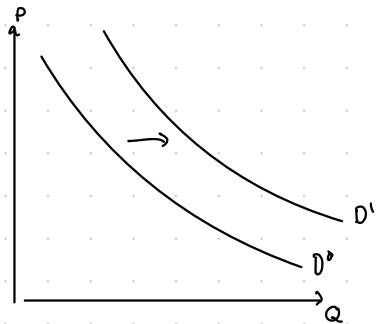
Suppose there is a change in DET.
↳ "shift" of demand.

(I) Income increases $\Delta I > 0$; $\Delta I = I_1 - I_0 > 0$

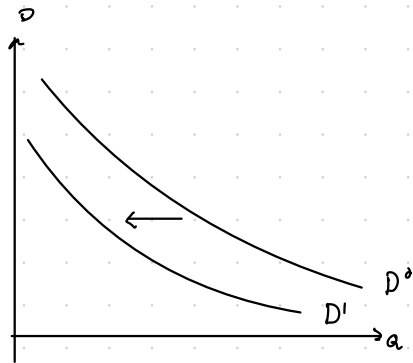
$\Delta I < 0$; $\Delta I = I_2 - I_0 < 0$



Price of Substitute $\uparrow$



Price of Complement $\uparrow$



Producers: the supply curve

The supply curve

The **supply curve** of a good or service tells us the total quantity of that good or service that sellers wish to sell at each price.

$$Q^s(p; \text{determinants})$$

Fixed set
of
determinants

$p \mapsto Q^s$

price
of
good

quantity
that is
offered
by producers

Determinants of supply — other factors that affect quantity supplied.

Examples:

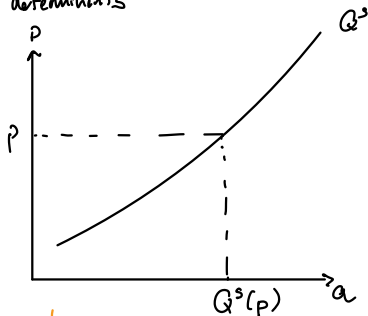
- Cost of inputs of production.
- # suppliers
- Technology
- Expectation of future prices
- Expectation of prices about inputs.

Graphical representation of the supply curve

$Q^s(p, \text{determinants})$

p - exogenous
 Q^s - endogenous.

given a fixed level of
determinants



Graph $\hookrightarrow$ Inverse supply:
 $(Q^s)^{-1}(q) \mapsto p$ (minimum price
that producers are
willing to accept)

Why do you think slopes upward?

Because of increasing O.C. in production

Law of Supply.

The supply curve are the $(Q^s(p), p)$
price-quantity pairs at which sellers are satisfied.

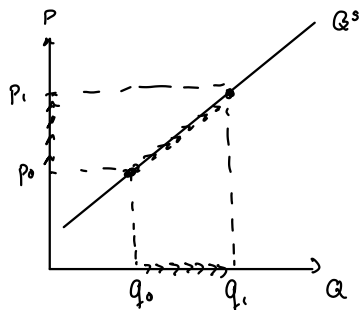
Horizontal interpretation: at price p , how much q^s
producers are willing to offer.

Vertical interpretation: at quantity q , what p
is the minimum price that
producers are willing to accept.

Movements along the supply curve and shifts of the supply schedule

Suppose DET are fixed

Consider a price change Δp

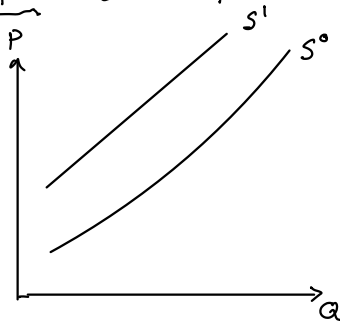


A $\Delta p \rightarrow$ movement
along the curve

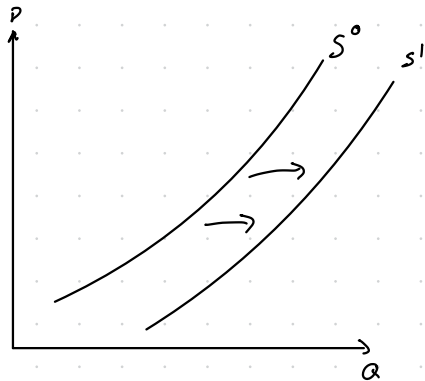
Suppose that DET changes

$\hookrightarrow$ Shift of curve

Example: Cost of inputs increase



There is technological progress $\Rightarrow$



Consumer Surplus

Consumer Surplus is the excess of the benefit received from a good over the amount paid for it. We can calculate consumer surplus as the marginal benefit (or value) of a good minus its price, summed over the quantity bought.

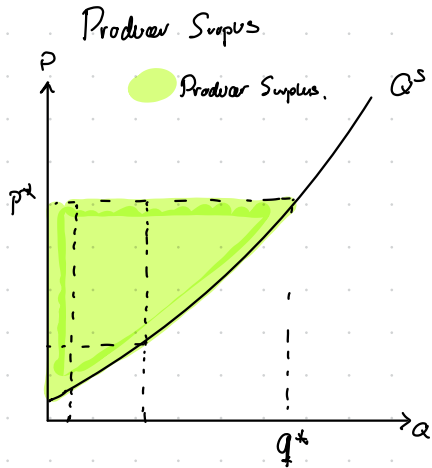
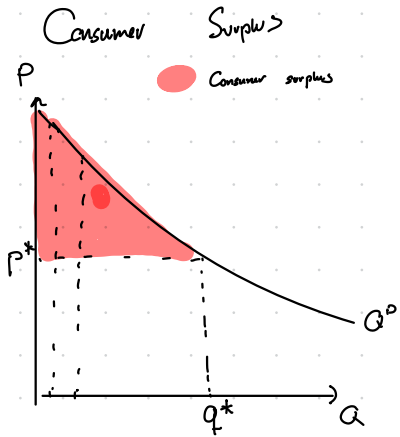
It is measured by the area under the demand curve and above the price paid, up to the quantity bought.

Producer Surplus

Producer Surplus is the excess of the amount received from the sale of a good over the cost of producing it.

We calculate it as the price received for a good minus the minimum-supply price (marginal cost), summed over the quantity sold.

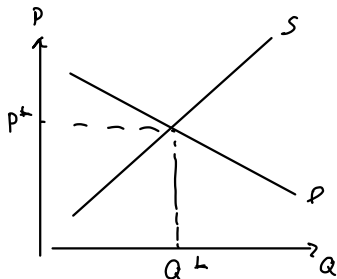
On a graph, producer surplus is shown by the area below price received and above the supply curve, summed over the quantity sold.



Market Equilibrium price and Quantity

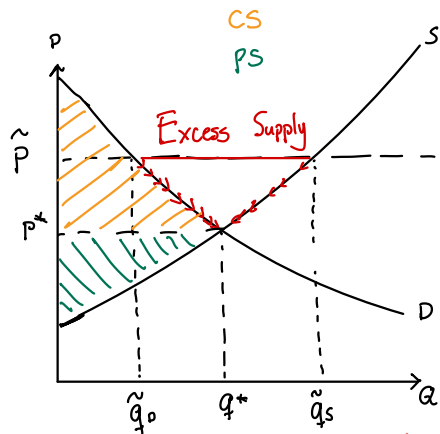
Equilibrium occurs at the price-quantity pair for which both buyers and sellers are satisfied.

$$(Q^*, P^*) \quad Q^D(P^*) = Q^S(P^*) = Q^*$$

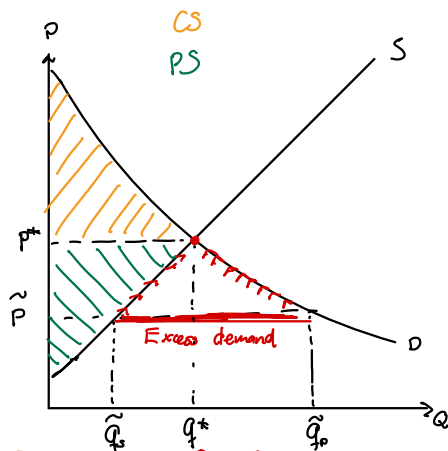


↓ Solve the system of eqs.
 $Q^D = Q^D(P)$
 $Q^S = Q^S(P)$ to find
 (Q^*, P^*)

Excess Supply and Excess Demand



$$\text{Excess Supply } (\tilde{P}) = \tilde{Q}_S - \tilde{Q}_D$$

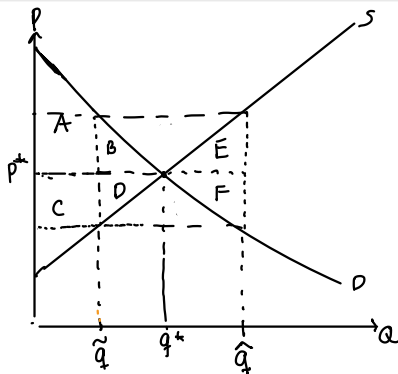


$$\text{Excess demand } (\tilde{P}) = \tilde{Q}_D - \tilde{Q}_S$$

- 1) The market undergoes a self-adjustment process: it forces the price to change to reach equilibrium
- 2) At the market equilibrium, Social Surplus (CS + PS) is maximized!

Social Surplus

Social surplus is the sum of producer and consumer surplus. It represents the total gains from participating in a market.



Suppose $\tilde{q}$ is traded at price p^*

$$CS = A$$

$$PS = C$$

What happens at $\hat{q}$ and p^* ?

$$CS = A + B$$

$$PS = C + D$$

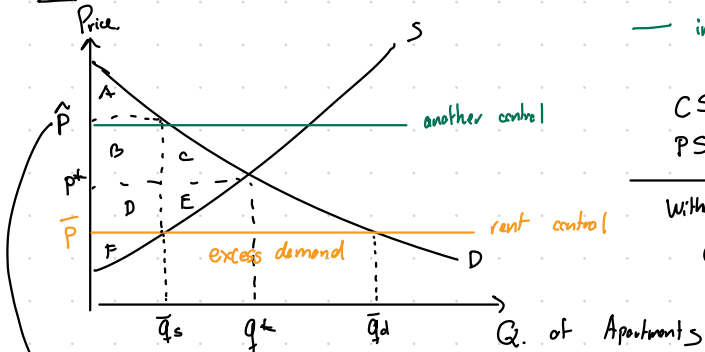
Suppose $\hat{q}$ is traded at price p^*

$$CS = A + B - F$$

$$PS = C + D - E$$

What happens if we impose restrictions?
 ↳ regulated

Ex: Rent Control.



↳ producers decide
 what happens: the offer $\bar{q}_s$ at price $\bar{p}$

rent control: price cannot go above $\bar{p}$
 — ineffective rent control.

$$CS = A + B + D$$

$$PS = F$$

Without the rent control

$$CS = A + B + C$$

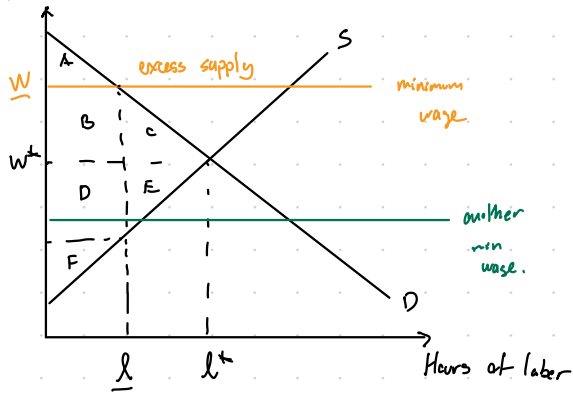
$$PS = D + E + F$$

What is lost?

$$DWL = C + E$$

Dead-weight Loss

w (hourly wage)



Labour Market

W is the minimum wage.
(price floor)

$$CS = A$$

$$PS = B + D + F$$

Without the restriction,

$$CS = A + B + C$$

$$PS = D + E + F$$

$$DWL = C + E$$

— ineffective min wage.

The Equilibrium Principle

A market in equilibrium leaves no unexploited opportunities for individuals, but may not exploit all gains achievable through collective action.

it maximizes
social surplus

↳ For later
OR

Political Economy course !!

Linear demand and supply curves

$$Q_D = a - bP$$

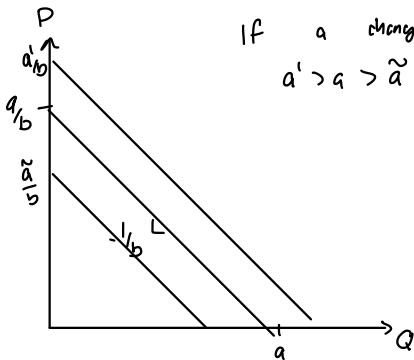
$$b > 0, a > 0$$

↓ To graph.
↓ find inverse demand
(solve for P)

$$P = \frac{a}{b} - \frac{1}{b} Q_D$$

a is a parameter encapsulating all demand determinants under consideration.

If a changes $\rightarrow$
 $a' > a > \tilde{a}$



Linear Supply

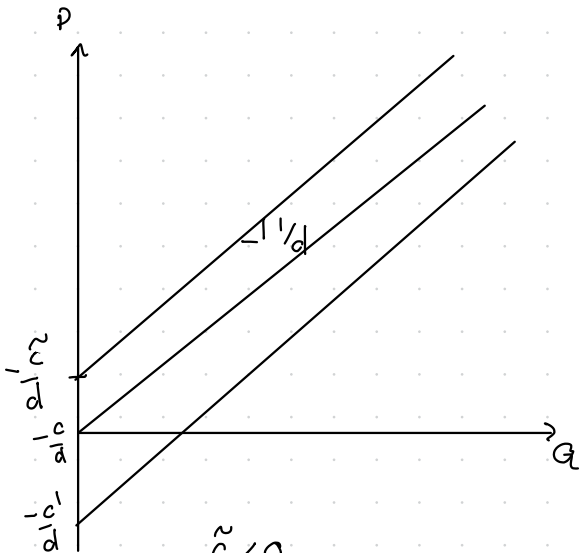
$$Q_s = c + dP$$

$$c \in \mathbb{R} \quad d > 0$$

↓ To graph it
find
inverse supply
(solve for P)

$$P = \frac{Q_s}{d} - \frac{c}{d}$$

c encapsulates
all Determinants
under consideration



$$\tilde{c} < 0$$

$$c^1 > 0$$

$$\bar{c} = 0$$

Equilibrium with linear supply and demand curves

$$Q_D = a - bP$$

$$Q_S = c + dP$$

For equilibrium:

$$Q_D(P^*) = Q_S(P^*)$$

$$a - bP^* = c + dP^*$$

$$a - c = (b + d)P^*$$

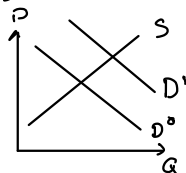
$$P^* = \frac{a - c}{b + d}$$

$$\begin{aligned} Q^* &= a - bP^* \\ &= a - b\left(\frac{a - c}{b + d}\right) \\ &= \frac{a(b + d) - b(a - c)}{b + d} \end{aligned}$$

$$Q^* = \frac{ad + bc}{b + d}$$

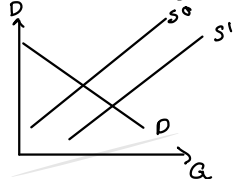
b, d are positive
 $a > 0, \quad c \in \mathbb{R}$

Shift of D ($a \uparrow$)



$P^* \uparrow$
 $Q^* \uparrow$

Shift of S ($c \uparrow$)



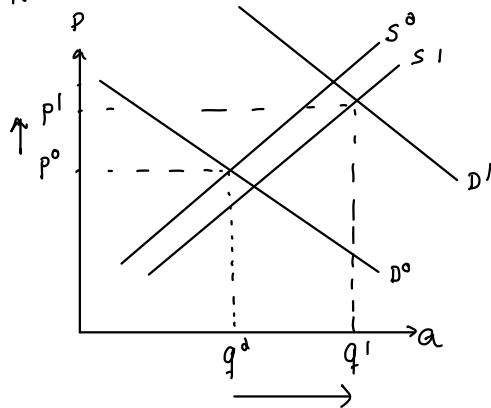
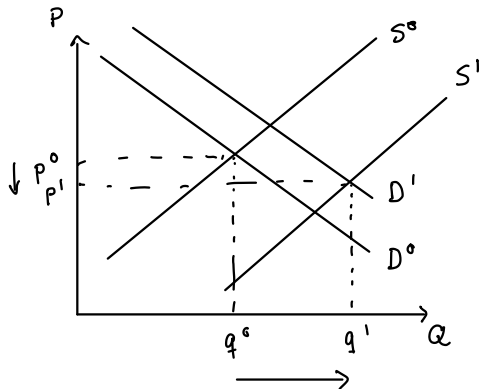
$P^* \downarrow$
 $Q^* \uparrow$

Comparative Statics Analysis (Ceteris Paribus Analysis)

What happens if both a and c change?

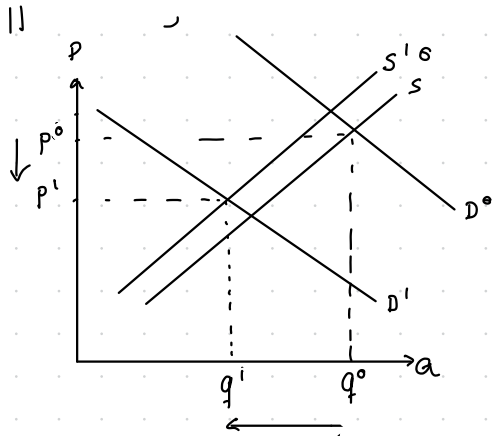
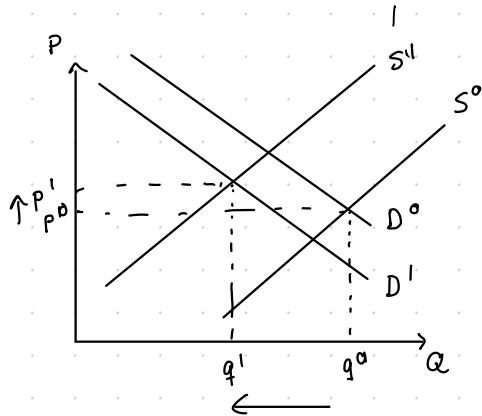
$$p^* = \frac{a-c}{b+d} \quad Q^* = \frac{ad+bc}{b+d}$$

, suppose $a \uparrow, c \uparrow$



Example: $a \downarrow$ $c \downarrow$ $p^* = \frac{a-c}{b+d}$ $Q^* = \frac{ad+bc}{b+d}$

HW: $a \uparrow, c \downarrow$
 $a \downarrow, c \uparrow$



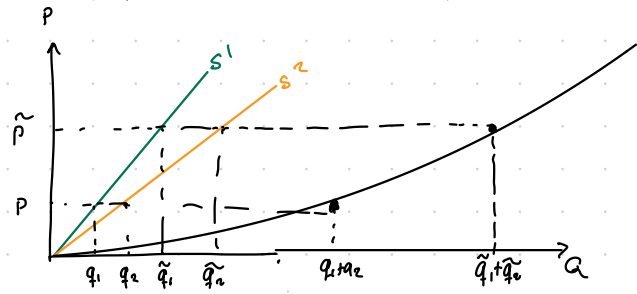
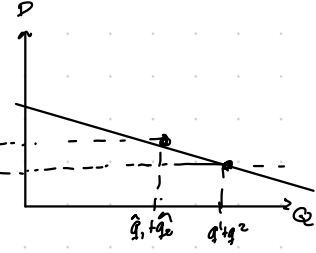
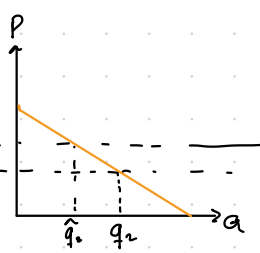
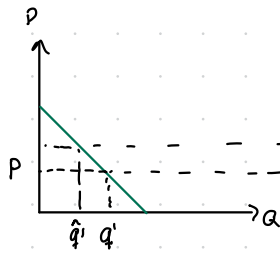
Aggregate Demand and Supply

Suppose that individual demand curves are given by $q_i^d(p)$, for $i = 1, \dots, I$ consumers; and individual supply curves are given by $q_k^s(p)$, for $k = 1, \dots, K$ producers/firms. The **aggregate or market demand** is obtained by **horizontal summation** of all the individual demand curves— at price p , we add up all demanded quantities:

$$Q^D(p) = \sum_{i=1}^I q_i^d(p).$$

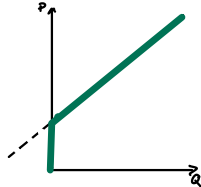
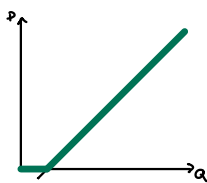
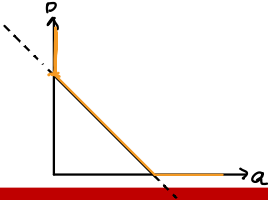
Similarly, the **aggregate or market supply** is obtained by **horizontal summation** of all the individual supply curves—at price p , we add up all supplied quantities:

$$Q^S(p) = \sum_{k=1}^K q_k^s(p).$$



$$Q_D: \underset{P}{\mathbb{R}_+} \longrightarrow \underset{Q_D}{\mathbb{R}_+}$$

$$Q_S: \underset{P}{\mathbb{R}_+} \longrightarrow \underset{Q_S}{\mathbb{R}_+}$$



Warning!

When doing the horizontal summation of the individual demand and supply curves to obtain the aggregate curves, you must always be careful to add them up at prices that lead to **positive quantities**! Remember, individual demand and supply curves are only defined for positive prices and positive quantities!

Examples

Suppose we have identical demands for all i , N of them.

$$q_i(p) = q(p)$$

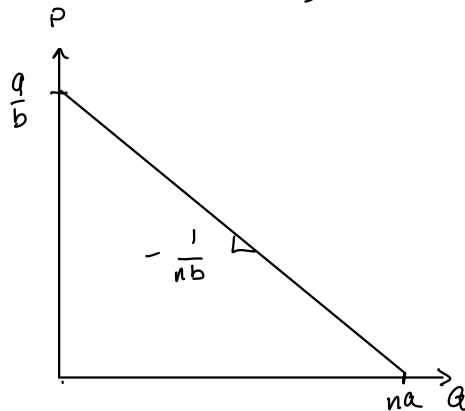
$$Q_D(p) = n q(p)$$

say $q(p) = a - bp$

$$Q_D(p) = na - nbp$$

$$na - Q_D = nbp$$

$$\frac{nq}{nb} - \frac{1}{nb} Q_D = p$$

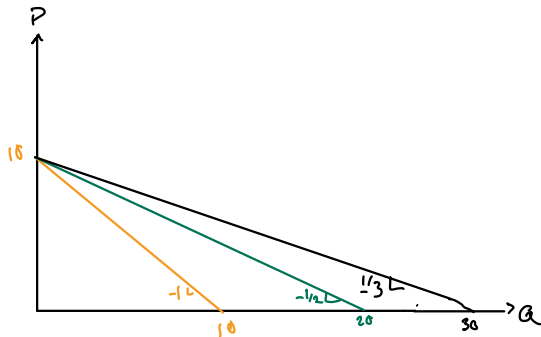


Examples

$$\underline{q_1(p)} = 10 - p$$

$$\underline{q_2(p)} = 20 - 2p$$

$$Q_0(p) = 30 - 3p = q_1(p) + q_2(p)$$



Examples

$$\underline{q_1(p)} = 1430 - 55p ; p = 26 - \frac{1}{55} q_1 \text{ Domestic } p$$

$$\underline{q_2(p)} = 1470 - 70p ; p = 21 - \frac{1}{70} q_2 \text{ World}$$

$$2900 - 125p$$

$$q_1(p) \geq 0 \Leftrightarrow 26 \geq p \geq 0$$

$$q_2(p) \geq 0 \Leftrightarrow 21 \geq p \geq 0$$

$$Q_p(p) = \begin{cases} 1430 - 55p & 21 \leq p \leq 26 \\ 2900 - 125p & 0 \leq p < 21 \end{cases}$$

